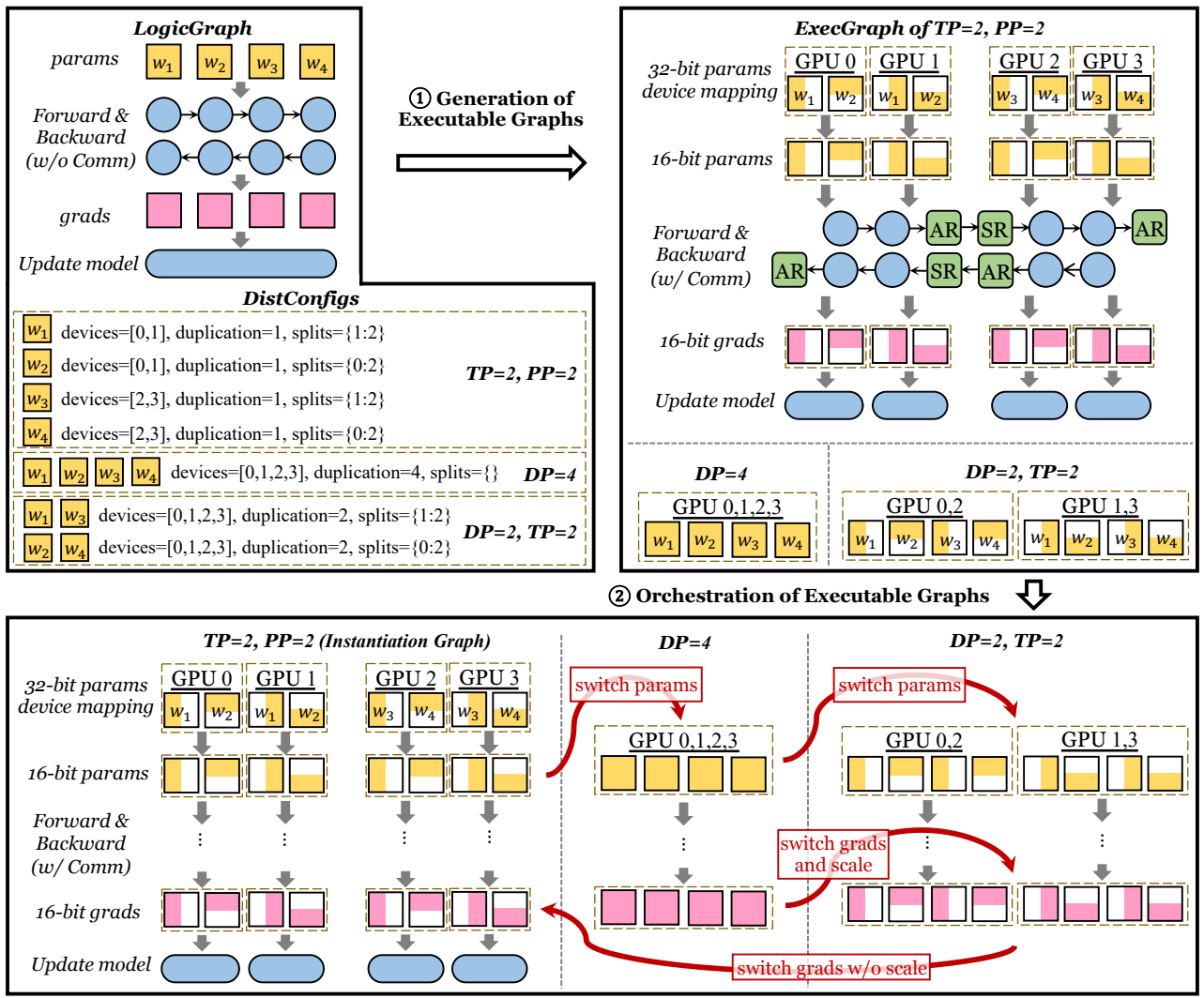

LogicGraph
params
Forward & Backward (w/o Comm)
grads
Update model
① Generation of Executable Graphs
DistConfigs
devices=[0,1], duplication=1, splits={1:2}
devices=[0,1], duplication=1, splits={0:2}
devices=[2,3], duplication=1, splits={1:2}
devices=[2,3], duplication=1, splits={0:2}
TP=2, PP=2
devices=[0,1,2,3], duplication=4, splits={}
DP=4
devices=[0,1,2,3], duplication=2, splits={1:2}
devices=[0,1,2,3], duplication=2, splits={0:2}
DP=2, TP=2
ExecGraph of TP=2, PP=2
32-bit params device mapping
GPU 0
GPU 1
GPU 2
GPU 3
16-bit params
Forward & Backward (w/ Comm)
AR
SR
16-bit grads
Update model
DP=4
GPU 0,1,2,3
DP=2, TP=2
GPU 0,2
GPU 1,3
② Orchestration of Executable Graphs
TP=2, PP=2 (Instantiation Graph)
switch params
switch grads and scale
switch grads w/o scale